University of Engineering & Technology

Department of Building Engineering & Construction Management

B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2021

Math 2123

(Mathematics-III)

Full Marks: 210 Time: 3 hrs

N.B. i) Answer any three questions from each section in separate script.

ii) Figures in the right margin indicate full marks.

Section – A

1. (a) A particle moves along the path $r = 2\cos t\,\hat{i} + 2\sin t\,\hat{j} + \hat{k}$ (35)
   (i) Find velocity, tangent vector, Normal vector and Binormal vector at $t = 1$.
   (ii) Find radius of curvature and torsion at $t = 0$
   (iii) Find osculating plane at $t = 1$.
   (iv) Also sketch the path and comment about it.

2. (a) Classify the vector field $\frac{\bar{r}}{r^2}$ by using differential operator '∇'. (10)

   (b) Find the angle of the two surfaces $x^2 + y^2 + z^2 = 12$ and $xyz = 8$ at (2, 2, 2). Also find the tangential plane to the surface $x^2 + y^2 + z^2 = 12$ at that point. Also find the directional derivatives of the scalar point function $f(x, y, z) = xyz - 8$ along x axis at that point. (15)

   (c) Find the workdone in a force field given by $\bar{F} = 3xy\hat{\imath} - 5z\hat{\jmath} + 10x\hat{k}$ along the curve $x = t^2 + 1, y = 2t^2, z = t^3$ from $t = 0$ to $t = 3$. (10)

3. (a) Evaluate $\iint_s \bar{q}.\hat{n}dS$ where S is a closed surface bounded by the cylinder $x^2 + y^2 = 9,\ z = 0$ to $z = 2,\ x = 0,\ y = 0$ and $\bar{q} = 4yx^2\hat{\imath} + 2y^2z\hat{\jmath} - z^2\hat{k}$. (15)

   (b) Find the volume of the region bounded by the planes $2x + 3y + z = 1,\ x = 0,\ y = 0,\ z = 0$. (10)

   (c) Test linear dependent and independent of vector $\bar{r}_1 = 3\hat{\imath} + \hat{\jmath} - 2\hat{k},\ \bar{r}_2 = \hat{\imath} + 3\hat{\jmath} + \hat{k}$ and $\bar{r}_3 = \hat{\imath} - 5\hat{\jmath}$. Also, if possible, express the dependent vector as a linear combination of independent vectors. (10)

4. (a) Define with examples: (i) Skew – symmetric matrix (ii) Singular matrix (iii) Diagonal matrix. (12)

   (b) Find the matrix A and the matrix B, such that $3A - 2B = \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix}$ and $-4A + B = \begin{bmatrix} -1 & 2 \\ -4 & 3 \end{bmatrix}$. (10)

   (c) When does a matrix have its inverse? If possible find the inverse of $\begin{bmatrix} 2 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$ (13)

Section – B

5. (a) Define the Laplace transform. Find the Laplace transform of the following functions: (14)

(i) $F(t) = \begin{cases} 3t, & 0 < t < 2 \\ 6, & 2 < t < 4 \end{cases}$, Where F(t) has period 4.

(ii) $F(t) = 3t^4 + 4e^{-3t} + t^2 \cos 2t$

(b) Find the Laplace transform of $\sin\sqrt{t}$. Also find $L\left\{\frac{\cos\sqrt{t}}{\sqrt{t}}\right\}$. (11)

(c) If $F(t) = 5\sin 3\left(t - \frac{\pi}{4}\right)$ for $t > \frac{\pi}{4}$ and 0 for $t < \frac{\pi}{4}$, find $L(F(t))$. (10)

6. (a) Find the inverse Laplace transform of $\frac{s+5}{(s+1)(s^2+4)}$. (12)

(b) Define the inverse Laplace transform. Find $L^{-1}\left\{\frac{3s+1}{(s-1)(s^2+1)}\right\}$ by using the Heaviside expansion formula. (12)

(c) Find the inverse Laplace transform of (i) $\frac{s^2+3}{s(s^2+9)}$ (ii) $\log\frac{s^2-1}{s^2}$ (11)

7. (a) Using Convolution Theorem find $L^{-1}\left\{\frac{s^2}{(s^2+1)(s^2+2)}\right\}$ (14)

(b) Solve the following differential equation by using the Laplace transform: $y'' - 3y' + 2y = e^{2t}$; $y(0) = -3$, $y'(0) = 5$. (10)

(c) Find the eigen values and eigen vectors for the matrix, $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ (18)

8. (a) Solve the system of equations,

$$x_1 - 2x_2 + 3x_3 - 4x_4 = 5$$
$$x_1 - 3x_2 + 7x_3 - 6x_4 = 7$$
$$2x_1 - 5x_2 + 10x_3 - 11x_4 = 12$$

(17)

(b) Find all non-trivial solutions of

$$3x_1 + 5x_2 - 2x_3 = 0$$
$$x_1 - 3x_2 + 4x_3 = 0$$
$$2x_1 + x_2 + x_3 = 0$$

$\frac{s}{s^2+1} \cdot \frac{s}{s^2+2}$

$\int_0^t \cos\sqrt{2}v \cos(t-v)\,dv$

$= \frac{1}{2}\int_0^t \cos(\sqrt{2}v - t + v) + \cos(\sqrt{2}v + v - t)\,dv$

$= \frac{1}{2}\left(\frac{\sin(\sqrt{2}v - v + t)}{\sqrt{2}-1} + \frac{\sin(\sqrt{2}v + v - t)}{\sqrt{2}+1}\right)_0^t$

$= \frac{1}{2}\left(\frac{\sin\sqrt{2}t}{\sqrt{2}-1} + \frac{\sin\sqrt{2}t}{\sqrt{2}+1} - \frac{\sin t}{\sqrt{2}-1} + \frac{\sin t}{\sqrt{2}+1}\right)$

$= \frac{1}{2}(2\sqrt{2}\sin\sqrt{2}t - 2\sin t)$

$= \sqrt{2}\sin\sqrt{2}t - \sin t$

$\begin{vmatrix} 3 & 5 & -2 \\ 1 & -3 & 4 \\ 2 & 1 & 1 \end{vmatrix}$

$= 3(-7) - 5(-7) - 2(7)$

$= 0$

So, trivial value / same

$\begin{bmatrix} 3 & 5 & -2 \\ 1 & -3 & 4 \\ 2 & 1 & 1 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$R_2' \to 3R_2 - R_1$, $R_3' \to 3R_3 - 2R_1$ $\begin{bmatrix} 3 & 5 & -2 \\ 0 & -14 & 14 \\ 0 & -7 & 7 \end{bmatrix}$

$R_3' \to 2R_3 - R_2$ $\begin{bmatrix} 3 & 5 & -2 \\ 0 & -14 & 14 \\ 0 & 0 & 0 \end{bmatrix}$

$3x_1 + 5x_2 - 2x_3 = 0$ $\quad x_3 = a$

$-x_2 + x_3 = 0$ $\quad x_2 = a$

$x_1 = -a$

$\therefore$ Soln $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = a\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$